ECET 251 Digital Communications
Wireless Link – Final Project Overview

Objectives

- Create a wireless system using a processor and RF modules
- Create a protocol for the system
- Test the communication system
- Document the system

Description

- This lab project will run 4 full lab periods but you should expect to spend some extra time outside of the lab to finish the project.
- Work alone or in groups and submit a (1) report for the group.
- The grade a group receives is based on the combined result of the work observed and demonstrated to their instructor, along with the project report submitted to D2L. Individual grades may differ, based on instructor's discretion.
- The major limitations/conditions are:
    - Project demo and report submission to D2L by Wednesday, April 8th
    - Send at least 8 bits of data wirelessly from one processing device to another
    - RF section must be a simple "transparent" radio which does no manipulation of the data
    - All data processing must be done by the computing device, not the RF modules
    - Output from the student's processor should be Manchester encoded and encompass the entire baseband which will be transmitted (may use another type of encoding at instructor's discretion).
    - Transmitter may not transmit for longer than 5 seconds at a time
    - Data must be variable driven (ie – read a switch, pot, rotary encoder, etc.) and it must perform a function (ie – turn on an LED, control motor speed, etc.)
    - Error detection is required in the receiving system. Error correction is a bonus.
    - Analysis of system RF characteristics must be included in the report:
        - Bandwidth & Spectrum use, Range limit, minimum Electric Field at Receiver to process data, Carrier frequency, Data Rate, Data Format, Operating Voltages, Power Levels, Error Rate, etc.

Report

The written report for this project will be in the form of a datasheet. The datasheet will describe the operation of the communication system. It should be written for someone that is interested in using your system for transmitting some data and has a basic understanding of electronics comparable to 1st year level. The datasheet is expected to be formatted properly (ie – formal), and include appropriate spelling and grammar.

The datasheet should include:

- An appropriate title page indicating the course, instructor, date of submission, group members, and a photo of your system
- A description of the wireless system and what it does (Block diagram and short paragraph)
- A description of the software used by the devices (Flowchart). This should include an indication of any software/code you have obtained from other sources. You do not need to include your code.
- A data table indicating: Bandwidth & Spectrum use, Range limit, minimum Electric Field at Receiver to process data, Carrier frequency, Data Rate, Data Format, Operating Voltages, Power Levels, Error Rate, etc.
- A screen capture showing the Manchester-encoded baseband into the transmitter with notations indicating the start/stop sequences, data payload, checksum, etc.

Final Project Grading Sheet

Group Members: _____

Final Grade: _____ / 25 = _____%

Demo / Observation Grading – points as listed below during demo of project. **(_____ /15)**

- **Professionalism** ___ / 5 → Attendance, Comportment, Workstation Cleanliness, etc.
- **Demonstration** ___ / 10 → Variable driven system judged on the following criteria:

Circuit Design ___ / 1 → Clean layout, no spaghetti wires, etc.	**Hardware Usage** ___ / 2 → LEDs, motors, sensors, actuators, speakers, etc.
Software design ___ / 2 → Layout, comments, credit for "stolen" code, etc.	**Functionality** ___ / 5 → No bugs, Error detection, etc.

Report Grading – points as listed below for D2L Submission. **(_____ /10)**

- **Presentation** ___ / 1→ Title Page, Grammar, Spelling, Font, Formatting, etc.
- **Describe the System** ___ / 5→ Explanation of system operation judged on following criteria:

Block Diagram ___ / 1 → System layout & explanation of features	**Flow Chart of SW Processes** ___ / 1 → Explanation of variable-driven code
Equipment List ___ / 1 → Detailed list of all components w/ specs.	**Summarizing Paragraph(s)** ___ / 1 → Written explanation of system
RF Characteristics ___ / 1 → Detailed summary of RF specs of your system (Freqs, Voltages, Power Levels, BW, etc)	

- **Analysis of System** ___ / 4→ Quantitative analysis of observed system characteristics:

EIRP @ Transmitter ___ / 1 → Data & Calculations	**Min. Electric Field @ Receiver** ___ / 1 → Data & calculations
Bit Error Rate ___ / 1 → Data & Calculations	**Manchester Encoded Signal** ___ / 1 → Scope capture & Notations